

Energy-saving behavior of urban residents in China: A multi-agent simulation

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ABSTRACT

With recent improvements in residents' quality of life and the implementation of the two-child policy, guiding the energy-saving behavior of urban residents has become a focus for achieving the national goals of the sustainable development strategy in China. Considering the subjective initiatives of individuals in a realistic environment is the key to studying energy-saving behavior and guiding policy making. This study builds a simulation model of the energy-saving behavior of urban residents using agent-based modeling and simulation (ABMS) methods, which are based on complex adaptive theory. By means of artificial neural networks and the Netlogo simulation platform, the subsequent effect of behavioral outcomes due to the short- and long-term influence of energy-saving behavior and intentions is analyzed in different policy situation. The results show that energy-saving intentions and behavior are poorly matched in the absence of an external policy framework. In the optimal policy situation, residents' energy-saving intentions and behavior have improved significantly. Policies can significantly encourage energy-saving intentions to become behavior. Different kinds of situational factors have different effects on intentions and the four types of energy-saving behavior. Finally, relevant policy implications are proposed based on analysis of the simulation results.

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1. Introduction

In the most highly populated country in the world, the energy consumption of Chinese residents has become an important source of energy demand and carbon emissions (Wang et al., 2015; Zhou and Yang, 2016; Liao and Shi, 2018). In recent years, the direct growth of the energy consumption of Chinese residents for daily living activities has reached 542.09 million tons of standard coal in 2016, accounting for 12.44% of total energy consumption and representing an increase of 1.97 times that in 2005 (NBSC, 2017). In addition, the study shows that energy consumption of families with children is generally higher than that of other families (Yue et al., 2013). After "the universal two-child policy" was instituted, the energy consumption of residents for daily living activities is expected to show a rapid growth trend. Therefore, guiding residents

to save energy and establishing a long-term mechanism to do so have become increasingly crucial. Energy demand-side management policy is considered a suitable way to realize sustainable development and pursue efficiency and fairness. At the same time, the concept of green consumption should be widely recognized, such as purchasing green building preferably (Song et al., 2018). A thrifty, green low-carbon, civilized, and healthy lifestyle and its associated consumption patterns should be formed, especially under the background of the sharp increasing in energy consumption for the past few years (Zhang et al., 2019). Hence, an in-depth study of residents' energy-saving behavior is needed.

As a kind of environmental protection behavior, the selection process of residents' energy-saving behavior is complex and has many influencing factors. Many related studies have been carried out on psychological characteristics, demographic and family characteristics, policy factors, and so on. In society, individuals not only interact with the environment, but are also affected by others. Therefore, under adjustments of situational factors, the purpose of this study is to construct a simulation model system representing residents' energy-saving behavior, realize the intelligence and interaction of resident subjects, and design the basic attributes,

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behavior rules, and simulation flow of various simulation subjects. The next step is to develop a simulation system to study the relationships between policy guidance and the emergence of energy-saving behavior and to evaluate dynamic adjustments to policy situation factors. At the same time, by comparing and analyzing the effect of different policy situational factors on different types of energy-saving behavior, the effect of behavioral results on long-term energy-saving intentions can be determined and the effectiveness of policy guidance can be explored, so as to make suggestions for guiding the design of policy systems.

This paper is organized as follows. Section 2 presents the literature review. Section 3 describes the construction of the conceptual model. Section 4 presents the agent attributes, behavior rules, and Netlogo system setting. In Section 5, the results of the analysis are presented and discussion is presented in Section 6. Finally, in Section 7, conclusions are drawn and policy implications are proposed.

2. Literature review

The existing literatures on energy-saving behavior mainly focus on influencing factors, guidance strategies, and energy-saving policies, among which the studies on influencing factors mainly focus on demographic factors, cognitive psychological factors, and external situational factors.

2.1. Demographic factors

Demographic factors mainly include gender, age, income level, and level of education and so on. There is no definite conclusion about the effect of gender on energy-saving behavior. Some studies have found that women are more likely than men to engage in pro-environmental activities, but others have concluded the reverse (Gaspar and Antunes, 2011; Vicente-Molina et al., 2013; Trotta, 2018). Frederiks et al. (2015) believed that age have a positive influence on individuals' energy-saving behavior, whereas Sardonou (2007) believed that this influence is negative. In addition, Belaid and Garcia (2016) have pointed out that energy-saving potential shows an inverted U-shape with age. Poruschi and Ambrey (2016) have showed that income level affects people's choice of energy-saving approach. High-income groups will make investments to improve energy efficiency, whereas low-income groups tend to adopt daily direct energy-saving measures. Education and knowledge of environmental issues clearly influence pro-environmental behavior (Vicente-Molina et al., 2013; Varela-Candamio et al., 2018; Zhang et al., 2018).

2.2. Cognitive psychological factors

Cognitive psychological factors mainly include values, attitude, subjective norms, and behavioral intentions. Researchers usually hold that values are closely related to environmental behavior (Wang et al., 2013; Zhang et al., 2018). The stronger the household environmental values that individuals or households have, the more possible it is for them to adopt energy-saving behavior. Many scholars regard attitude as an important and positive psychological variable affecting energy-saving behavior (Ajzen and Driver, 1991; Ek and Söderholm, 2010; Andrzej et al., 2014; Prete et al., 2017; Taufique and Vaithianathan, 2018; Casaló and Escario, 2018). Liu et al. (2018) hold that residents' subjective cognition, perceived usefulness and environmental attitude could predict their willingness to buy green buildings. Based on descriptive analysis and path analysis, Wang et al. (2013) have revealed that behavioral intention plays the most important role in explaining the level of sustainable consumption behavior. Wang et al. (2016) have surveyed 443 urban residents in China and found that the intention to

purchase hybrid electric vehicles significantly affects purchasing behavior.

2.3. External situational factors

As individuals in society and a behavior subject, residents can feel social role effects and social pressure in the process of adapting to the surrounding environment and gradually form some cognition of social values and moral norms through learning or imitation, thus affecting their behavior. Therefore, external situational factors such as social pressure, policy, and the behaviors of other individuals' also have an important influence on an individual's choice of behavior (Ajzen and Driver, 1991; Nowak et al., 2014). Nolan et al. (2008) have conducted an energy-saving survey of 810 residents in California and found that social norms are more predictive of energy-saving behavior than other related beliefs. Handgraaf et al. (2013) have argued that social incentives can encourage people to abide by social norms and hence engage in more green consumption behavior. Pro-environmental social norms have a significant impact on both green purchasing and green behavior (Liobikienė et al., 2016).

According to the Theory of Planned Behavior, if the behavioral agents have stronger subjective norms, they will also have stronger behavioral intentions (Ajzen and Driver, 1991). As in society, energy-saving individuals not only interact with the natural environment, but are also influenced by other people's behavior. Individuals have the characteristics of social learning, social comparison, and social identity. Through observing and comparing other people's behavior and social norms, individuals will adjust their self-perception and evaluation to adopt energy consumption patterns that are consistent with others or socially identifiable (Ajzen and Driver, 1991; Mi et al., 2018). Researchers often use intervention experimental methods to explore the impact of different intervention strategies, including information feedback, goal commitment, role model demonstrations, and peer education, on energy-saving behavior. A study by Winett et al. (1985) shows that role model demonstrations can promote social learning, enabling observers to make energy-saving behavior consistent with role models. Komatsu and Nishio (2015) have found that social comparison has a significant positive impact on energy-saving behavior. Through field experiments, Nilsson et al. (2015) have found that group employees who received feedback have a higher level of changing energy-using behavior than those who do not. Fujimi et al. (2016) further found that customized information works best for energy-saving behavior when used in conjunction with public commitments.

A few scientists have used simulation methods to explore the impact of external situational factors on energy-saving behavior. Anderson et al. (2014) constructed a model to study how energy consumption behavior spreads through social networks and described how households change their energy consumption behavior through social influence. Furthermore, Anderson and Lee (2016) constructed an empirical behavior model to simulate the effect of information feedback on energy-saving behavior. The results showed that when individual or normative feedback was sent to all users in the network, total energy consumption tended to reach an average value. Lopes et al. (2016) used the Energy Plus/Design Builder software as a simulation tool to estimate the potential of various energy-saving behaviors. The results indicated that daily energy-saving behaviors and investing in efficient equipment both had significant energy-saving potential, but that investing in efficient equipment had higher energy-saving potential than daily energy-saving behavior.

From the existing literatures, it is not hard to discover that studies on demographic and cognitive psychological factors often

use questionnaire survey methods, structural equation models, and multivariate statistical analysis. Other scientists have used intervention methods to study external situational factors. However, these methods cannot easily clarify the dynamic adjustment of situational factors and compare the impact on various types of energy-saving behavior. Using these methods, it is also hard to determine the long-term effects of behavioral consequences on energy saving and to evaluate the impact of interaction between residents on their behavioral intentions.

Therefore, under the adjustment of situational factors, the objective of this study was to study the inconsistency between individual intention and behavior and the impact of this inconsistency on the perception of individual behavior results and on subsequent behavioral intentions. Whether dynamic interaction among individuals leads to dynamic changes in behavioral intentions is also discussed.

3. Conceptual model

3.1. The basis of model construction

To construct the conceptual model and obtain basic data on energy-saving behavior and the influencing factors of urban residents, a questionnaire survey was conducted. The questionnaire was based on a study of the energy-saving behavior of urban residents in Jiangsu Province, which was carried out in previous research by the authors (Yue et al., 2013). The questionnaire contained social demographic characteristics, psychological characteristics, energy-saving behavior, behavioral outcome perceptions, and policy situational factors. In November 2013, a total of 653 paper questionnaires and 867 online questionnaires were collected from urban residents in eastern China. Among these, there were 520 valid paper questionnaires and 683 valid online questionnaires, and the effective recovery rate was 79.14%. The results of reliability and validity analysis showed that Cronbach's alpha coefficient for each of the variables was above 0.6 and that the correlation coefficient between the items and the total was greater than 0.4. The scales of the Kaiser-Meyer-Olkin (KMO) and Bartlett test results and of exploratory and confirmatory factor analysis results were reasonable. Therefore, the reliability and structural validity of the scale were deemed to be high. The results of descriptive statistical analysis showed that effective samples were evenly distributed over the social-demographic characteristics (gender, age, educational background, and occupation type), as well as over the distribution of household statistical characteristics (residential types, residential area, number of family members, household structure, household income level, and the proportion of household energy expenditure). Afterwards, the inherent and external influencing factors and the mechanism of the energy-saving behavior were analyzed by means of statistical analysis software and the structural equation model, which provided the empirical basis and data foundation for building the simulation model.

In this paper, the energy-saving behavior of urban residents was divided into four types according to a literature review and previous interviews with residents. (1) Habit adjustment behavior (HAB) refers to the behavior of reducing energy consumption by changing daily energy use habits without sacrificing quality of life. For example, residents can turn off the lights when leaving the room or unplug the power supply to reduce energy consumption when electrical appliances are not used for a long time. (2) Quality threshold behavior (QTB) refers to behavior that reduces energy consumption by changing patterns of daily energy use, thus sacrificing a certain quality of life. For example, residents can purchase electrical appliances less or use air conditioning and heaters in a controlled manner. (3) Energy efficiency investment behavior (EIB)

refers to the reduction of energy consumption by investing in energy-efficient products or equipment to improve energy efficiency. For example, residents take the initiative in purchasing energy-saving lamps and efficient household appliances. (4) Interpersonal facilitation of behavior (IFB) refers to a situation where residents have not only committed themselves to saving energy, but also encourage others to implement energy-saving behavior through interpersonal activities. For example, residents take the initiative to convey energy-saving information to friends or colleagues, share energy-saving experiences, participate in energy-saving and environmental protection communities or organizations, and prevent other people's energy waste.

The outcome perception of energy-saving behavior refers to how residents perceive the benefits of behavior change after implementing energy-saving behavior. This paper divides this topic into two dimensions: the outcome perception of economic savings and that of spiritual satisfaction.

By analysis of the questionnaire data using multivariate statistical and structural equation models, it was found that situational factors have a regulatory effect on the process leading from behavioral intention to behavior implementation, i.e., energy-saving behavior is influenced both by an energy-saving intention and by external situational factors which refer to the external factors that affect the environmental behavior of individuals (Ajzen and Driver, 1991; Andrzej et al., 2014). Individuals' perception of the behavioral outcome can have an impact on behavioral intentions, and therefore the individual determines whether to implement energy-saving behavior based on an energy-saving behavioral intention and external situational factors (Iwata et al., 2015; Ajzen, 2002). According to the results of the questionnaire survey, external situational factors have a regulatory effect on the process leading from energy-saving behavioral intention to behavior. Among external situational factors, social norms related to energy saving (SNES), popularization of economic energy-saving policies (PEEP), popularization of guided energy-saving policies (PGEP), execution and validity of energy-saving policies (EVEP), execution and validity of information intervention (EVII), and price level of energy (PRIE) play a positive role in moving behavioral intention towards behavior implementation. Energy-saving product attributes (EPA) only have a positive effect on investment behavior regarding energy-efficient products, and the price of energy-saving products (PRIP) has a negative effect on them (as shown in Table 1).

3.2. Model construction

From a microcosmic perspective, the specific behavior of agents, the parameters, the interaction rules among agents, and the interaction rules between agents and the environment were set based on complex adaptive system (CAS) theory and the ABMS method. The system was simulated by combining the microscopic behavior of the system and the macroscopic "emerging" phenomena (Taylor and Suzana, 2018). Therefore, with the help of CAS theory and agent simulation modeling theory, the conceptual model of the energy-saving behavior of urban residents was designed on the basis of an empirical analysis of the questionnaire survey.

In the simulation model of urban residents' energy-saving behavior, it was assumed that the urban residents' energy-saving management system is a dynamic and integral system composed of a plurality of agents with nonlinear interaction according to the initialization rules. The simulation system consisted of urban resident agents, energy-saving policy-maker and enforcer agents, and management agents, among whom various interactions occurred. The management agents collected information about residents'

Table 1
Abbreviations and definitions of variables.

Variables	Abbreviations	Definitions
Energy-saving behavior of urban residents	Habit adjustment behavior	HAB
	Quality threshold behavior	QTB
	Energy efficiency investment behavior	EIB
External situational factors	Interpersonal facilitation behavior	IFB
	Social norms related to energy saving	SNES
	Popularization of economic energy-saving policies	PEEP
	Popularization of guided energy-saving policies	PGEF
	Execution and validity of energy-saving policies	EVEP
	Execution and validity of information intervention	EVII
	Price level of energy	PRIE
	Energy-efficient product attributes	EPA
	Price of energy-efficient products	PRIP

energy-saving behavior through observation, evaluated the policy effect, and gave feedback to the policy-making agents. Then the policy-making agents adjusted the energy-saving policy measures and the process of transforming behavioral intention into behavior implementation according to the feedback. The interaction of urban residents with others can also change urban residents' energy-saving intention and behavior (as shown in Fig. 1).

It was assumed that an energy-saving behavioral intention is an expectation of energy-saving outcomes. After implementing energy-saving behavior, urban residents determine the degree of consistency between the energy-saving behavioral results and their expectations. If there is a difference between the two, the intention will then be gradually adjusted. The individual also has the ability

to communicate within the social sphere. The individual's intention and behavior are influenced by the interaction between the individual and the external environment. In addition, the individual is also affected by interaction with other residents, based on their own subjective norms regarding what they expect to achieve in the interaction.

4. Methods

The Netlogo simulation system developed in this study included an artificial neural network subsystem and a behavior simulation subsystem. Dynamic adjustment and intervention of system parameters was realized by integrating the artificial neural network

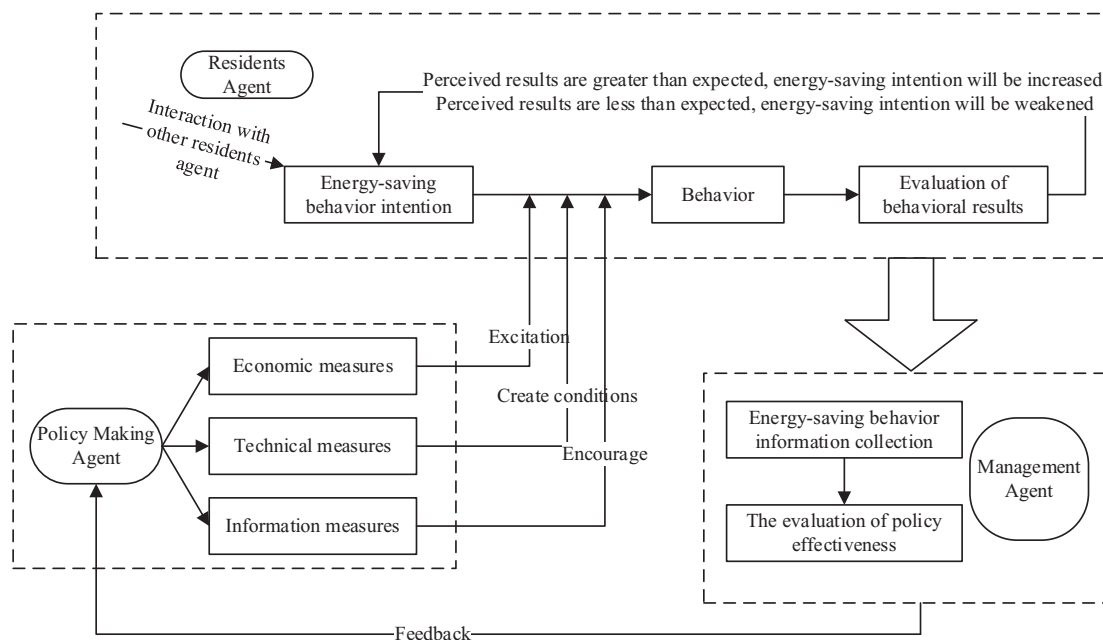


Fig. 1. Simulation concept model based on ABMS method.

subsystem and the behavior simulation subsystem into a model. The simulation system operation had three main stages: initialization, artificial neural network training, and energy-saving behavior simulation. In the initialization stage, the "Setup" command was used to initialize the simulation system. During the artificial network training stage, the developed artificial neural network was trained according to the survey data until the computational accuracy of the network converged within an acceptable range. In the energy-saving behavior simulation stage, the behavioral simulation was carried out according to the given parameters, which were controlled. The changes in the output results under different control parameters were observed, and the results were compared with the normal state. A detailed description of the simulation system design is given in the following section.

4.1. Agent attributes and behavior rules setting

In the absence of interference from external factors, behavior is entirely determined by intention. Therefore, it can be assumed that the behavioral intention of residents is defined to be the expectation of outcome when implementing the corresponding energy-saving behavior. The residents' intention to save energy is determined by their psychological characteristics. The psychological characteristics are endogenous and have some stability in the short term. In this situation, expectations and behavioral results are equal, and the intention to save energy will not change. If residents perceive the results of energy-saving behavior to be greater than their expectations, the intention to save energy will slowly increase; if they are less than expected, the intention will slowly weaken.

The attributes and behavioral rules of the resident agents in the Netlogo platform were set as follows: (1) The number of resident agents. The number of resident agents is a controllable parameter that can be set freely. To make the model error smaller and the degree of fit better, 500 research data points were randomly selected from 1203 research data points as the basic data of the artificial neural network, and the number of resident agents was set to 500. (2) Location and activity range of resident agents. According to the number of resident agents, resident agents were randomly assigned in the system, and their range of activities was randomly fixed. This study assumed that the size of the Netlogo world was 49×41 tiles, i.e., the number of tiles was 2009. The range of activities was the result of the total number of tiles divided by the total number of resident agents. (3) The interactive range of each resident agent. The interaction between resident agents was completed by daily interaction with the surrounding residents. It was assumed that the activity range of each resident agent was the set of tiles located less than 2 units from its coordinates, as shown in Fig. 2. Within this scope, the resident agents could carry out their interactions in accordance with their own subjective norms, and their energy-saving intentions could be adjusted accordingly. (4) At time instant T , the energy-saving intention of the resident agent was determined by the initial energy-saving intention and the results of the initial energy-saving behavior. (5) At time instant T , the energy-saving behavior was determined by the energy-saving behavioral intention and the external environment change at instant $T-1$. The specific action path was mainly determined by the artificial neural network. Table 1 shows the specific settings of each resident agent's attributes, such as subjective norms, energy-saving behavioral intentions, energy-saving behavior, and the results of energy-saving behavior.

According to the survey results, each respondent had the natural properties of strong or weak subjective norms. It was assumed that

the probability of the subject's intention to adjust behavior was determined by the value of the subjective normative attribute. The subjective norms (SN_j) of the resident agent were set to a stochastic and continuous value between 0 and 1, representing subjective norms from the weakest to the strongest. Behavioral intentions ranged from those that were most likely to be affected by other agents to those that would not be affected by other agents. Each resident agent compared the subjective norm with the agents whose peripheral distance was less than 2 units away. The bigger the difference, the greater the probability of intention adjustment. In addition, it was assumed that each agent had no difference, and the initial intention attribute was randomly assigned. The assignment method was to extract a value randomly from "1, 2, 3, 4, 5", which indicated the initial value of the energy-saving intention of the resident agent. "1" indicated that residents were reluctant to save energy, "2" that residents were less willing to save energy, "3" that residents were neutral, "4" that residents were more willing to save energy, and "5" that residents were very willing to save energy.

In the interaction between the resident agent and the external environment, it was assumed that a change in the external environment was consistent with the behavior of each agent in the system. A resident agent's perception of external situational factors will affect the choice of energy-saving behavior by residents, with the resident agent gradually adjusting behavior until a stable intention and behavior were formed. Residents' energy-saving behavior can be expressed as a function of energy-saving behavioral intention and situational factors. The outcome perception of the behavioral subject was determined by intention, behavior, and situational factors. Therefore, the residents' behavior and behavioral perception functions can be calculated by the artificial neural network, which contains the intention and situational factors of energy-saving behavior, based on the data from the questionnaire survey.

4.2. Netlogo system setting and parameter design based on the artificial neural network

According to the statistical analysis results of the survey data and the conceptual simulation model, an artificial neural network was constructed based on the survey data. The input variables were subjective norms, the initial values of energy-saving behavioral intentions, and policy situational factor variables. The output variables were energy-saving behavioral intentions, outcome perceptions of energy-saving behavior, and four kinds of energy-saving behavior. For each kind of behavior, "1" indicated the minimum value of each context factor and "5" indicated its maximum value. The context factor value could be varied continuously from 1 to 5. Therefore, a back-propagation (BP) artificial neural network could be established to calculate the energy-saving behavior and the behavioral results for each resident agent. The input had 10 nodes, including the subjective norms, the initial values of energy-saving behavioral intentions, and eight kinds of policy situational factor variables. The output had 6 nodes, including energy-saving behavioral intention, results of energy-saving behavior, and four kinds of energy-saving behavior. The number of hidden layer nodes is usually obtained by empirical and trial-and-error methods. To simplify the model, the hidden layer nodes were set to 1. By repeated trial and error, it was discovered that when the number of hidden layer nodes was set to 20, the error convergence of the neural network model was the fastest. At the same time, the learning rate was set to 0.010. After a period of learning, the error of the network was reduced to 0.0465, and the convergence of the neural network was satisfactory, which indicated that the actual survey data and the artificial neural network were well fitted (as

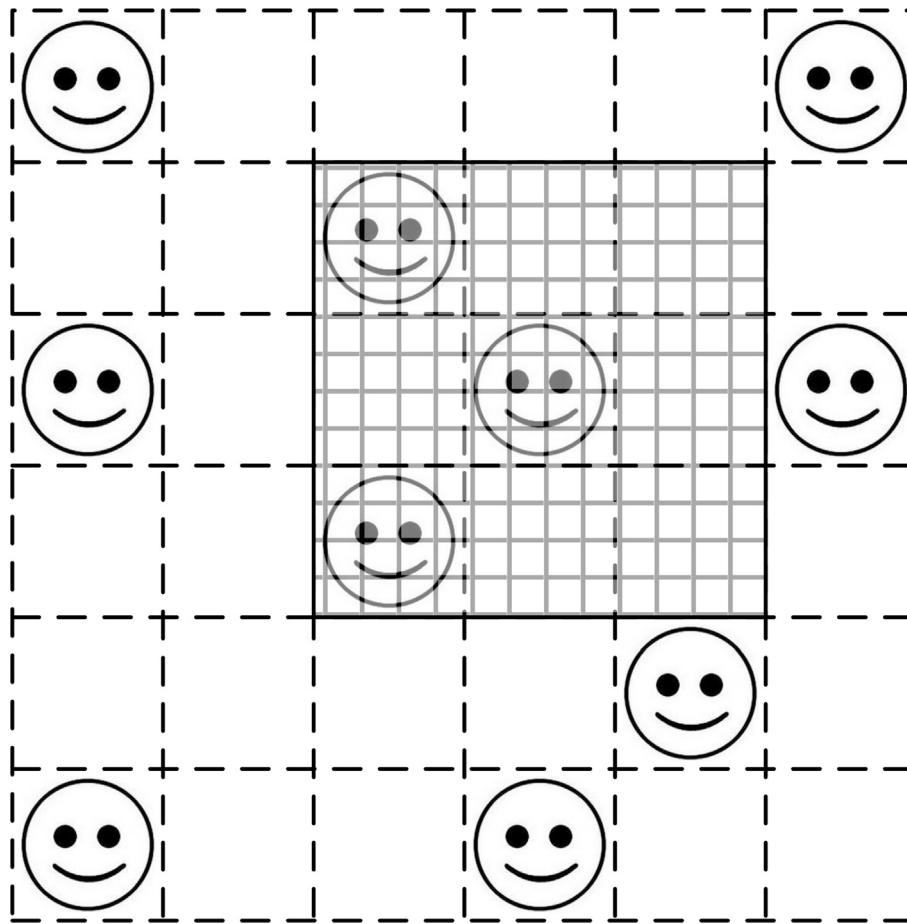


Fig. 2. Distribution of resident agents on a two-dimensional lattice.

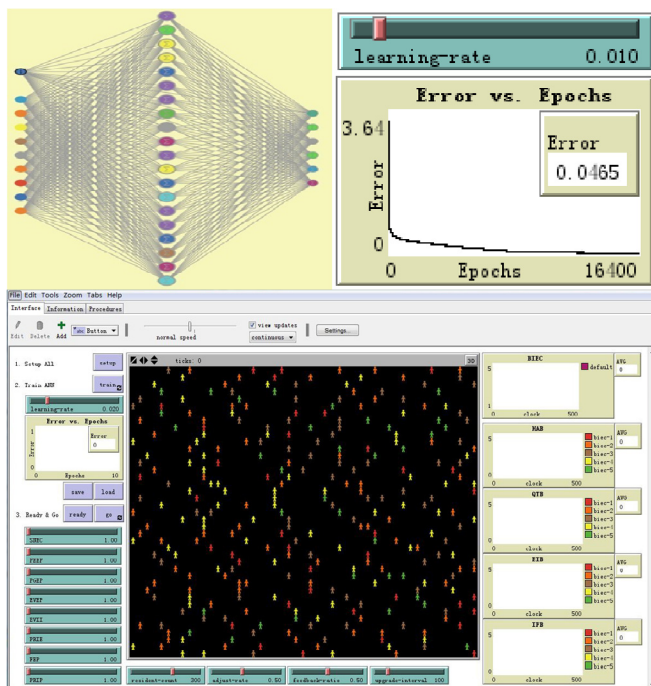


Fig. 3. Artificial neural network and convergence curve.

shown in Fig. 3).

The following parameters were also included in the simulation system: (1) The adjustment rate of behavior and intention. To simplify the model and increase its running speed, the adjustment rate of behavior and intention was set to 0.2, i.e., the adjustment value for each Tick intention was 20% of the remaining value (the difference between expectation and results) of the last Tick intention. (2) The feedback ratio of intention change, which was used to control weights in terms of the effects of two behavioral outcomes (economic savings and spiritual satisfaction) on intention change. According to a regression analysis of the survey data, the long-term and slow contribution rates of economic-saving perception and spiritual-satisfaction perception to energy-saving behavioral intention were 20% and 80% respectively. It was assumed that a given change in behavior will always result in the same proportional adjustment of energy-saving behavioral intention, but the adjustment cycle was relatively long. (3) The updated interval of intention, which was used to control the length of the cycle in which the resident agent could adjust its intention to complete the change according to external environment and behavioral choices. It was assumed that the cycle of intention change was 200, i.e., an agent needed 200 Ticks to complete the intention adjustment. The parameter settings of the adjustment rates of behavior and intention and the update interval of intention did not have a substantial effect on the simulation results and were designed to make simulation results easier to obtain.

Table 2
Attribute-settings for resident agents.

Attributes of resident agent	Attribute description
Subjective norms of agent	SN_j
The initial value of the energy-saving behavior intention of agent	$I_j^0 = f(e_1, e_2, \dots, e_{11})$
HAB of agent at the time T	$B_{j,H}^t = f_1(I_j^t, SNEC_j^t, PEEP_j^t, PGEP_j^t, EVEP_j^t, EVII_j^t, PRIE_j^t)$
QTB of agent at the time T	$B_{j,Q}^t = f_2(I_j^t, SNEC_j^t, PEEP_j^t, PGEP_j^t, EVEP_j^t, EVII_j^t, PRIE_j^t)$
EIB of agent at the time T	$B_{j,E}^t = f_3(I_j^t, SNEC_j^t, PEEP_j^t, PGEP_j^t, EVEP_j^t, EVII_j^t, PRIE_j^t, FEP_j^t, PRIP_j^t)$
IFB of agent at the time T	$B_{j,F}^t = f_4(I_j^t, SNEC_j^t, PEEP_j^t, PGEP_j^t, EVEP_j^t, EVII_j^t, PRIE_j^t)$
The perception of economic savings outcome of agent at the time T	$PE_j^t = f_5(I_j^t, SNEC_j^t, PEEP_j^t, PGEP_j^t, EVEP_j^t, EVII_j^t, PRIE_j^t, FEP_j^t, PRIP_j^t)$
The perception of spiritual satisfaction outcome of agent at the time T	$PS_j^t = f_6(I_j^t, SNEC_j^t, PEEP_j^t, PGEP_j^t, EVEP_j^t, EVII_j^t, PRIE_j^t, FEP_j^t, PRIP_j^t)$
Energy-saving behavior intention of agent at the time T	$I_j^t = f(I_j^0, PE, PS)$

Note: SN represents subjective norms; j represents the j-th resident agent; I represents the resident agent's intention to save energy; and $e_1, e_2, \dots, e_{11}$ represents factors of individual psychological characteristics. The superscript 0 indicates the initial period, and t represents the t-th period. B represents the resident agent's choice of energy-saving behavior. The subscript H represents HAB. The subscript Q represents QTB. The subscript E represents EIB. The subscript F represents IFB. PE represents the perception of economic savings outcome from the implementation of energy-saving behavior. PS represents the perception of spiritual satisfaction outcome from the implementation of energy-saving behavior.

5. Results

5.1. Setting of the simulation mode of policy situation

The simulation system was divided into benchmark mode and analysis mode in this study. The benchmark mode was based on the statistical mean values of policy situational factors. Among these, SNES was 2.50, PEEP was 3.28, PGEP was 2.19, EVEP was 2.28, EVII was 3.08, PRIE was 3.40, EPA was 3.54, and PRIP was 4.26 (as shown in Table 3). Different residents chose the energy-saving behavior appropriate to them and perceived behavioral results based on their behavioral intention. The state of the residents' environment was stable, and the values of their intentions did not change in the benchmark mode.

The analysis model included the policy-free incentive mode, the optimal policy mode, and the mode for the implementation of the various policies alone. In the analysis mode, the residents' cognition of situational factors was changed by adjusting the external environment. Then the behavioral choices of residents were changed and their perception of the behavioral results changed accordingly. Residents slowly adjusted their intention to adopt energy-saving behavior according to the degree of matching between the results of the behavior and their expectations. The change tendencies of residents' behavioral intentions and behavioral choices were further observed and analyzed under environment change. According to empirical analysis, SNES, PEEP, PGEP, EVEP, EVII, and PRIE had a positive effect on energy-saving behavior. EPA had a positive effect on EIB, and PRIP had a negative effect on EIB. To better compare and analyze the effects of different policy factors, it was assumed that PRIP had a maximum value of 5 and the other situational factors had a minimum of 1, which was defined as the policy-free incentive mode. This mode was assumed to be the initial state of the system analysis mode. In contrast to the policy-free incentive mode, the policy optimal mode set the situational factor with positive adjustments to the maximum value of 5, and the situational factor was set with negative adjustments to the minimum value of 1.

The modes of various policies acting alone included modes of SNEC, PEEP, PGEP, EVEP, EVII, PRIE, EPA, and PRIP. The mode setting method served to adjust one of the policy values to the maximum value and to fix the remaining policy values at the minimum value. For example, the SNEC mode served to adjust the value of SNEC

from 1 to 5, leaving the other situational factors fixed at their initial state values. The PRIP mode adjusted the value of PRIP from 5 to 1, leaving the other situational factors fixed at their initial state values. Among these, because the energy-saving product attributes and energy-saving product price levels only affect EIB, only the output results of energy-saving behavioral intention and the EIB were included in the energy-saving product attributes and energy-saving product price level mode.

5.2. Simulation output results in different policy situation modes

In the benchmark mode, according to the calculation results of the artificial neural network, the behavioral choices, the behavioral results, and the energy-saving behavioral intentions of resident agents were trained to achieve an essentially stable state. Fig. 4 shows the simulation output results of behavioral intentions and the four kinds of energy-saving behavior. The output diagrams for the other policy modes were similar to Fig. 4, and hence the results have been summarized in Table 2 to better compare the output results.

From Fig. 4 and Table 4, the following conclusions can be drawn from the simulation output results for the various parameter-setting modes:

- (1) In the benchmark mode, the stable outputs of the system are basically consistent with the empirical research results, indicating that the random sample means were basically no difference from the means of large sample. Therefore, it can be considered that the basic data of artificial neural network was selected reasonably.
- (2) In the policy-free mode, the resident agents have low values of behavior intention of energy saving and energy-saving behavior. The residents who have stronger intention also implement HAB and EIB, but the output values are lower than the ones in the benchmark mode. The resident agents with different initial behavior intention of energy saving seldom implement QTB and IFB without policy incentives. Comparing the policy-free mode to the benchmark one, residents' behavior intention of energy saving is poorly matched with all kinds of energy-saving behavior.
- (3) Under the optimal policy mode, the output values of four types of energy-saving behavior and behavior intention of

Table 3
Descriptive statistical analysis results of moderated variables.

Variables	Mean values	Standard deviation	Item	Mean values	Standard deviation
SNES	2.50	0.895	SNES1	2.80	0.857
			SNES2	2.54	0.893
			SNES3	2.34	0.923
			SNES4	2.33	0.906
PEEP	3.28	1.424	PEEP1	3.57	1.501
			PEEP2	3.51	1.284
			PEEP3	3.41	1.412
			PEEP4	2.93	1.453
			PEEP5	2.99	1.471
PGEP	2.19	1.352	PGEP1	2.26	1.349
			PGEP2	2.12	1.355
EVEP	2.28	1.370	EVEP1	2.04	1.420
			EVEP2	2.52	1.320
EVII	3.08	0.983	EVII1	3.19	0.993
			EVII2	2.97	0.973
PRIE	3.40	0.751	PRIE1	3.38	0.764
			PRIE2	3.41	0.738
EPA	3.54	0.952	EPA1	3.85	0.881
			EPA2	3.38	1.019
			EPA3	3.51	1.003
			EPA4	3.43	0.904
PRIP	4.26	0.682	PRIP1	4.26	0.682

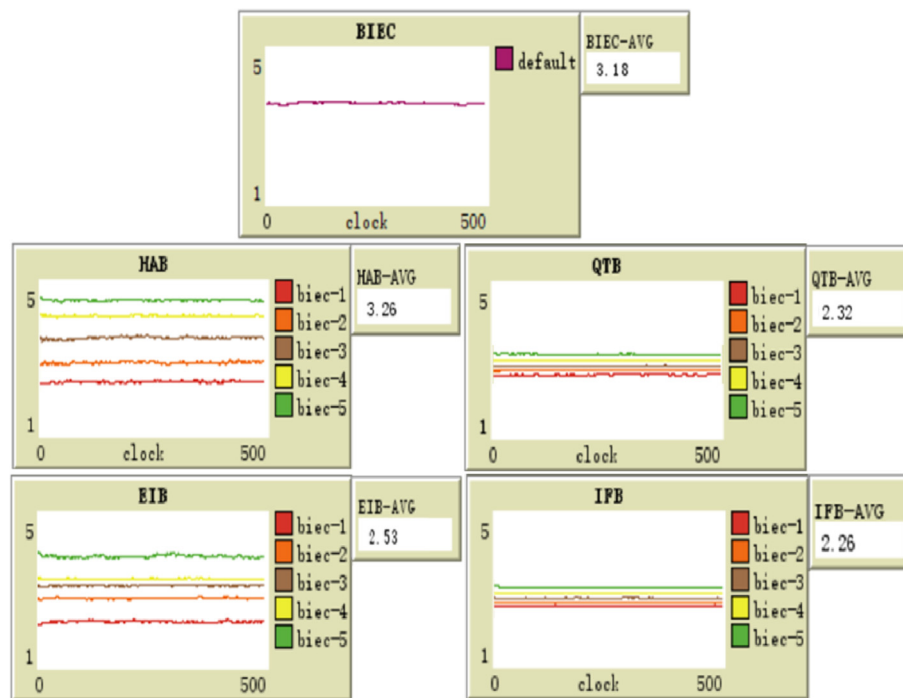


Fig. 4. Results of system output in the benchmark mode.

energy saving of resident agents have significantly and sharply increased. The output values of the simulation system are stable after running for a period of time. The mean value of HAB increases from 1.61 to 2.03 to 4.41, the mean value of QTB increases from 1.61 to 4.18, the mean value of EIB increases from 1.94 to 4.84, and the mean value of IFB increases from 1.62 to 4.1. The mean value of behavior intention of energy saving increased from 2.84 to 3.30. Therefore, in the external optimal state, EIB will be best implemented, followed by HAB, and then QTB and IFB. From the difference between the final output and the initial state, among the four types of energy-saving behavior of resident agents, EIB

increases most sharply, followed by QTB, indicating that these two types of energy-saving behavior are most affected by external situational factors. In general, under the optimal policy mode, all kinds of energy-saving behavior increased significantly. The average values of all kinds of energy saving behaviors are relatively with high stable output values finally. The behavior intention of energy saving also increases slowly after the behavior changes, and the increase also is relatively significant.

- (4) In the mode of SNEC, the output values of four types of energy-saving behavior of resident agents all increase. The increase of HAB is most significant, increasing from the initial

Table 4

Simulation output results for different parameter-setting modes.

Policy situational modes of intention and behavior	Intention and behavior				
	Energy-saving behavior intention	HAB	QTB	EIB	IFB
The benchmark mode	3.18	3.26	2.32	2.53	2.26
The policy-free incentive mode	2.84	2.03	1.61	1.91	1.62
The optimal policy mode	3.3	4.41	4.18	4.84	4.1
The mode of SNEC	3.23	4.22	3.13	2.83	3.58
The mode of PEEP	2.88	3.77	3.93	3.97	1.8
The mode of PGEP	3.19	4.08	2.9	2.69	3.63
The mode of EVEP	3.1	4.02	3.36	3.53	3.28
The mode of EVII	3.02	3.27	3.08	3.35	2.89
The mode of PRIE	2.87	4.15	4.08	3.89	1.91
The mode of EPA	2.84	—	—	4.16	—
The mode of PRIP	2.86	—	—	4.27	—

2.03 to 4.22. IFB increases from 1.62 to 3.58 and QTB increases from 1.61 to 3.13. Meanwhile, EIB of resident agents increases weakly from 1.91 to 2.83. The behavior intention of energy saving increases from 2.84 to 3.23. Overall, the influence of SNEC to HAB is mostly remarkable, that is to say, SNEC can promote the residents' HAB significantly. The influence of SNEC to HAB is followed by the one to IFB and then QTB. It showed that awareness of energy saving of residents will be improved being influenced by SNEC. Thereby, residents will be more willing to sacrifice a certain quality of life to save energy.

- (5) As is shown in the mode of PEEP, the change of IFB output is small and increases from the original 1.62 to 1.80. HAB increases from the original 2.03 to 3.77, QTB increases from the original 1.61 to 3.93 and EIB increases from the original 1.94 to 3.97. The value of behavior intention of energy saving increases from the initial 2.84 to 2.88. In general, PEEP has no obvious effect on IFB. But PEEP can encourage residents to develop good habits of energy-saving behavior and implement EIB. Moreover, the higher understanding of economic policies, the more willing residents were to sacrifice certain quality of life to realize energy saving. However, PEEP has no significant effect on residents' behavior intention to save energy.
- (6) In the mode of PGEP, among resident agents' four types of energy-saving behavior, the change of EIB is small, increasing from the original 1.94 to 2.69. Unlike EIB, QTB increases from the original 1.61 to 2.90, HAB increases from the original 2.03 to 4.08 and IFB increases significantly from the original 1.62 to 3.63. Behavior intention of energy saving increases from the original 2.84 to 3.13. Overall, PGEP influences EIB weakly, but can better guide QTB, HAB and IFB. The more popularized energy-saving policies are, the more willing to sacrifice the quality of life to achieve energy saving the residents were. PGEP also can promote the residents to develop good habits of saving energy, and actively promote others to save energy. Behavior intention of energy saving also can be intensified for a long time.
- (7) Under the mode of EVEP, all four types of energy saving behavior of resident agents have increased. HAB increases from the original 2.03 to 4.02, QTB increases from the original 1.61 to 3.36, EIB changes from the original 1.94 to 3.53, and IFB increases from the original 1.62 to 3.28. Residents' behavior intention of energy saving increases from the original 2.84 to 3.10. In general, EVEP can better promote residents' HAB, EIB and IFB. The higher EVEP implementation is, the more willing the residents are to sacrifice a certain quality of life to achieve energy saving. EVEP also can

promote residents' behavior intention of energy saving in the long run.

- (8) It could be seen in the mode of EVII, the residents' four types of energy-saving behavior increases: HAB increases from the original 2.03 to 3.27, QTB increases from the original 1.61 to 3.08, EIB increases from the original 1.94 to 3.35 and IFB increases from the original 1.62 to 2.89. Behavior intention of energy saving changes from the original 2.84 to 3.02. As a whole, EVII better promotes residents to develop good energy-saving behaviors. The higher the degree of information intervention is and the more effective information intervention is, residents are more willing to sacrifice certain quality of life to realize energy saving and implement EIB and IFB. EVII has a long-term promotion to residents' behavioral intention of energy saving.
- (9) Under the mode of PRIE, IFB has no significant change, while the other three types of energy saving behaviors change obviously. HAB increases from 2.03 to 4.15, QTB increases from 1.61 to 4.08, EIB increases from 1.94 to 3.89. But residents' energy intention of energy saving only increases by 0.03 with no significant change. In general, the improvement of energy price level can better promote residents to adjust their behavior habits, sacrifice a certain quality of life to realize energy saving and invest on energy efficiency. But the impact on residents' behavior intention to save energy is not significant.
- (10) In the mode of EPA, EIB of the resident agents increases from 1.94 to 4.16, indicating that EPA has a significant impact on EIB. Under the mode of PRIP, EIB increases from 1.94 to 4.27 and 2.33, indicating that the price level of energy-efficient products has a significant impact on EIB of resident agents. Behavior intention of energy saving is not influenced by the EPA and PRIP significantly.

6. Discussion

A comparison of the simulation results of the policy-free situation and the optimal policy situation reveals that the energy-saving behavioral intentions and the QTB has the worst match when external policy factors do not have an incentive effect. This indicates that people who are willing to save energy are reluctant to sacrifice their quality of life and to implement QTB. The degree of matching between energy-saving behavioral intentions and IFB is also low. This indicates that even if residents have energy-saving behavioral intentions, most of them will not take the initiative to carry out QTB and IFB. Relatively speaking, residents have a high degree of matching between intention and HAB and EIB. However, judging from the mean values of the outputs of behavior and intention, the intention is not fully transformed into behavior. In

the optimal policy mode of external environment, all types of energy-saving behavior and energy-saving intentions significantly increase, which indicating that overall optimization of policy factors will have a significant impact on residents' energy-saving behavior and intentions.

Under the influence of different situational factors, SNES has the highest degree of influence on adjusting HAB. PRIE and PEEP have the highest degree of impact on QTB. EPA, PRIP, PEEP, and PRIE have higher impacts on EIB. PGEP and SNES have higher impacts on IFB. On the whole, SNES have a significant impact on HAB, QTB, and IFB.

PEEP has a significant impact on HAB, QTB, and EIB, among which EIB is the most strongly influenced. PGEP has significant impacts on HAB, QTB, and IFB, as well as on EI. EVEP can significantly promote the implementation of four types of energy-saving behavior and energy-saving behavioral intention. Increasing EVII can effectively promote the implementation of energy-saving behavior among residents, for example detailed bills of energy use and information intervention with energy-saving advice can encourage residents to save energy. PRIE has significant effects on HAB, QTB, and EIB. EPA and PRIP have significant impacts on EIB, but have little effect on energy-saving behavioral intentions.

From the impact of different situational factors on the same type of energy-saving behavior, HAB was mostly affected by SNES. QTB is mostly influenced by PRIE and PEEP. EIB is mostly affected by PRIP, EPA, and PEEP, followed by PRIE. IFB is mostly affected by PGEP, followed by SNES. PEEP and PRIE have the most significant impact on QTB and EIB, indicating that residents are willing to save energy through purchasing energy-efficient products and reducing the pursuit of life comfort under the incentive of economic measures. PGEP and SNES play significant roles in HAB and IFB.

Overall, SNES and PGEP have the greatest impact on the intention to save energy, followed by EVEP and EVII. The impacts of PGEP, SNES, EVEP, and EVII are more extensive and effective. They can both influence energy-saving intentions and behavior and stimulate the implementation of energy-saving behavior for a long time by strengthening energy-saving intentions. SNES has greater effects on HAB, QTB, and IFB. PGEP can significantly affect the implementation of HAB and IFB and stimulate residents to sacrifice a certain quality of life to save energy. EVEP and EVII have significant effects on all four types of energy-saving behavior. In the long run, these four kinds of policies can strengthen residents' intention to implement energy-saving behavior. Therefore, the effect of these four kinds of policy measures has universal and lasting characteristics compared with other types of policy measures. Therefore, it is the best policy group for guiding residents to implement energy-saving behavior over the long-term.

7. Conclusions and policy implications

7.1. Conclusions

Based on survey data, this study used CAS theory and artificial neural networks to simulate the energy-saving behavior of urban residents using the Netlogo multi-agent simulation platform and drew the following conclusions.

- (1) The choice process of residents when moving from energy-saving intentions to energy-saving behavior is a steady state formed by the long-term comprehensive effect of individual internal psychological factors and external environmental factors. Energy-saving intentions and behavior are poorly matched in the absence of an appropriate external policy. EI and QTB have the worst match, followed by IFB. The overall optimization of policies has significant impact on residents' energy-saving behavior and intentions.

- (2) There are differences in the effects of various situational factors on energy-saving intentions and the four types of behavior. The promotional effect of SNES is significant in promoting transformation from behavioral intention to HAB, EIB, and IFB. SNES can change the energy-saving concepts of residents on a spiritual level and can improve residents' energy-saving behavioral intentions in the long run. Among policy popularization factors, the degree of PEEP and PRIE has significant effects on the transformation of energy-saving behavioral intentions to HAB, QTB, and EIB. Price, tax, and other economic policy measures can induce residents to feel the pressure of expenses and hence to pay more attention to adjusting their energy habits and living comfort and to the energy efficiency of energy-consuming equipment. While residents feel energy consumption pressure and focus on energy efficiency at the same time, subsidies can further encourage residents to buy energy-efficient products. However, PEEP has little impact on residents' energy-saving behavioral intentions in the long term. It is also difficult to change energy-saving intentions at the spiritual level. This reflects the observation that economic policy can only stimulate energy-saving behavior. Once the policy disappears, the corresponding behavior will also disappear.
- (3) Policy situational factors at the spiritual level can motivate residents to adjust their energy-saving behavioral intentions. The behavioral results of economic savings have less effect on energy-saving intentions, but spiritual satisfaction can significantly affect the intention to save energy. In the long term, SNES and PGEP have the greatest impact on the intention to save energy, with EVEP and EVII also having great influence. After policy formulation, policy publication and implementation are also pivotal. The execution and validity of guided economic energy-saving policies can not only induce residents to adjust their energy-saving habits, but also can encourage residents to sacrifice a certain quality of life to achieve energy savings, carry out energy-efficiency investment, engage in interpersonal facilitation of energy-saving behavior, and enhance residents' energy-saving behavioral intentions over the long-term. Therefore, we should pay attention not only to the formulation and promotion of energy-saving policies, but also to their implementation and supervision, so that residents can understand the relevant content of the policy and make the corresponding behavioral changes.

7.2. Policy implications

In summary, creating good energy-saving social norms, improving the degree of popularization of guided energy-saving policies, and strengthening the execution and validity of policy implementation and information intervention can stimulate residents to take the initiative to induce others to implement energy-saving behavior, resulting in a good energy-saving interpersonal environment that will help residents develop good energy-saving habits and engage in a rational pursuit of quality of life. Moreover, these measures can change the energy-saving concepts of residents on the spiritual level and strengthen the intention behind residents' energy-saving behavior. This will encourage the implementation of energy-saving behavior over the long-term.

Interaction between residents has a great impact on their energy-saving behavior. We should pay attention to the impact of residents' subjective norms on energy-saving behavior. Therefore, the government should enrich their guided energy-saving policies, increase the intensity of education and publicity on energy-saving

behavior, create a good atmosphere of energy-saving social norms, and create better energy-saving communication platforms to strengthen the energy-saving behavioral intentions of residents. The government should strengthen the execution and validity of information intervention, encourage residents to develop energy-saving targets, build and improve the query platforms that provide instant information to households, strengthen the management of household energy consumption, and increase the corresponding publicity to encourage residents to inquire actively about household energy consumption information. Raising residents' awareness of their own energy consumption can stimulate initiatives to implement energy-saving behavior.

Author contributions

Conceptualization, T.Y. and R.L.; Data curation, T.Y., J.L. and H.L.; Formal Analysis, T.Y., J.L. and H.L.; Investigation and Simulation, T.Y., J.L. and H.L.; Methodology, T.Y.; Resources, T.Y., and R.L.; Validation, R.L. and H.C.; Writing-Original Draft Preparation, T.Y., J.L. and H.L.; Writing-Review & Editing, T.Y., R.L. and Y.G.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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